

# Evaluating the Effectiveness of ML Algorithms in Water Potability Prediction Aligned with SDGs

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**Abstract**—This research addresses the critical need for accurate water quality prediction, aligning with Clean Water and Sanitation (SDG 6) and Good Health and Well-being (SDG 3). Utilizing a diverse dataset of water quality attributes, we developed and evaluated multiple machine learning models to predict water potability, contributing to SDG 9 (Industry, Innovation, and Infrastructure) through technological advancement. Our primary contribution is a Hyperparameter-Tuned Histogram Gradient Boosting Classifier, which outperformed several other algorithms, achieving an accuracy of 67%, a ROC-AUC score of 70%, and an F1-score of 66%. The study highlights challenges of class imbalance and feature distribution in water quality prediction, addressing aspects of SDG 11 (Sustainable Cities and Communities) by improving urban water management. Our findings have significant implications for water resource management, supporting SDG 12 (Responsible Consumption and Production) through efficient resource utilization. While acknowledging limitations such as single dataset reliance, this research combines modern machine learning with comprehensive analysis to enhance water quality assessment. It contributes to SDG 17 (Partnerships for the Goals) by providing a framework for collaboration between technology and environmental sectors. Future directions emphasize improving model performance and real-world applicability, potentially supporting SDG 13 (Climate Action) by enhancing resilience to water-related climate impacts.

**Index Terms**—Multi-criteria Decision Analysis, Calibrated Random Forest, User Experience, consumer-centric ISP selection, data-driven & personalized ISP recommendations

## I. INTRODUCTION

Access to clean water is an basic human right for every individual. However water quality assurance is a major challenge for this contemporary world. As natural water sources are facing contamination due to climate change and technological revolution, the need for technology to detect and predict potable water efficiently is imperative. Moreover, Water quality is a critical global concern, intrinsically linked to several Sustainable Development Goals (SDGs), particularly SDG 6 and SDG 3 [1], [2]. The complex interplay of physical,

chemical, and biological factors determining water quality necessitates sophisticated assessment methods, especially for evaluating potability [3]. This imperative resonates across local and national levels, intertwining health imperatives with broader developmental paradigms aligned with SDG 11 (Sustainable Cities and Communities) [4], [5].

Recent studies have highlighted that investments in water supply and sanitation infrastructure can yield substantial net economic benefits, supporting SDG 8 (Decent Work and Economic Growth) [6], [7]. These gains materialize through the attenuation of adverse health effects and the reduction in healthcare expenditures, often surpassing the costs of implementing these interventions. In recent years, machine learning techniques have emerged as powerful tools for water quality prediction, outperforming models based solely on physical principles [8], [9]. This shift towards data-driven approaches aligns with SDG 9 (Industry, Innovation, and Infrastructure), promoting technological advancement in environmental management [10]–[12]. Our research focuses on a rigorous examination of various machine learning algorithms, aiming to develop a robust predictive model for water quality assessment.

The motivation behind this project stems from the critical importance of ensuring safe water resources for public health and sustainable development, as emphasized in SDG 12 (Responsible Consumption and Production) [13], [14]. Conventional models relying solely on physical principles often lead to suboptimal real-world predictions [15], [16], necessitating more advanced approaches.

Our study aims to comprehensively explore Machine Learning (ML) techniques for accurate water quality prediction and develop an efficient predictive model which will address the comparison of different ML tools to predict potable water based on various water attributes. This will exhibit how hyper parameter tuning intensify the accuracy level of ML models on imbalanced dataset. This aligns with SDG 17 (Partnerships for the Goals) by fostering collaboration between environmental

science and data analytics [17]. Additionally, by enhancing our ability to predict and manage water quality, this research indirectly supports SDG 13 (Climate Action), as climate change significantly impacts water resources [18].

The primary objectives of this study encompass the comprehensive exploration of ML techniques for accurate water quality prediction, followed by the development of a robust and efficient predictive model. The research aims to:

- 1) **Assessment of Machine Learning Techniques:** Thoroughly investigate and evaluate a diverse array of machine learning algorithms to determine their efficacy in predicting water quality, focusing on the potability of water for human consumption.
- 2) **Comparison and Selection of SOTA (state of the art) ML models:** Conduct a detailed comparison of the performance of different ML algorithms to identify the most accurate and reliable model for water quality prediction.
- 3) **Predictive Model Development:** Build a predictive model capable of accurately classifying water samples as potable or non-potable based on a set of relevant attributes, thereby enhancing the precision of water quality assessment.

The remaining paper is arranged in the following sections. Section II describes the methodology employed in this study. Followed by this section, Section III presents the experimental result found from this study and Section IV concludes this study with future directions.

## II. METHODOLOGY

Employed methodology in this study is divided into different steps namely, data acquisition and processing, assessment of existing machine learning models and a predictive model development. The steps of this methodology are described briefly in the later sections.

### A. Data Acquisition and Processing

Utilized dataset is sourced from Kaggle.com, titled “Water Quality - Drinking Water Potability” [19]. It encompasses 10 distinct attributes designed to facilitate the training and evaluation of our predictive model. The dataset classifies water into two categories: potable and non-potable. Comprising 3276 data points, all attributes are numerical in nature except the potability attribute which is categorical. Table I provides an overview of the dataset, detailing the attributes, brief descriptions, and data types.

Once the data are collected, they undergo a processing step to check for null values and duplicates. It was observed that the pH, Sulfate, and Trihalomethanes columns contained some null values. These null values were subsequently replaced with the mean of their respective columns. After that Synthetic Minority Over-sampling Technique (SMOTE) technique is used to balance the dataset. This approach using SMOTE ensures that both classes in the ‘Potability’ feature are represented equally, effectively addressing the initial class imbalance in the dataset. SMOTE achieves this by creating synthetic examples

of the minority class based on the feature similarity with existing minority class instances, thereby enhancing the overall balance and improving the performance of ML models trained on imbalanced datasets.

1) *Dataset Description:* Table II presents a comprehensive statistical summary of the dataset attributes, detailing key metrics such as the count, mean, standard deviation (std), minimum (min), 25th percentile (25%), median (50%), 75th percentile (75%), and maximum (max) values. The dataset comprises 3,276 observations and encompasses various water quality parameters, including pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. These metrics provide a foundational understanding of the distribution and variability of the dataset attributes, which are critical for subsequent analysis. The pH values range from 0 to 14, with a mean of 7.08, suggesting a neutral to slightly alkaline nature. Hardness has an average value of 196.37 mg/L and a standard deviation of 32.88, indicating variability in mineral content. The solids attribute shows a high mean of 22014.09 mg/L, reflecting significant levels of dissolved substances. Chloramines and sulfate have means of 7.12 mg/L and 333.79 mg/L, respectively, which are within expected ranges for treated water. Conductivity averages 426.21  $\mu$ S/cm, denoting ion concentration. Organic carbon has a mean of 14.28 mg/L, and trihalomethanes average 66.40 ppm, both aligning with safety guidelines. Turbidity is low, with an average of 3.97 NTU, well below the WHO limit. The dataset quality appears robust, with comprehensive attribute coverage and reasonable ranges.

In the dataset, the potability feature is distributed in a 61% to 39% ratio. Specifically, 61% of the data points are classified as non potable, while the remaining 39% are classified as potable. Numerical features’ distribution with respect to the potability feature is presented in Figure 1. From this distribution plot, we can clearly see the difference in the water classification. For example, distribution of non potable water is high on conductivity compared to potable water. Same observation is applicable to Turbidity, Trihalomethanes. However, Ph value, Chloramines, Sulfate, Organic carbon presence doesn’t show significant differences. The larger version of the figure can be found in <https://github.com/ping543f/icrest24-38/blob/main/feat-dist.png>

### B. Assessment of Machine Learning Models

After completing data processing, various ML models were evaluated using the processed dataset. This assessment included classic state-of-the-art models such as Random Forest (RF), Support Vector Machine (SVM), and 22 other models. Additionally, deep learning models, namely Perceptron, BiLSTM, and BiGRU, were employed for training and testing the dataset. Performance evaluation of these models was conducted based on metrics including Accuracy, ROC-AUC, and F1 scores.

TABLE I  
DATASET ATTRIBUTE DESCRIPTIONS

| Attribute       | Description                                                                   | Type        |
|-----------------|-------------------------------------------------------------------------------|-------------|
| pH Value        | Measures the acid-base balance of water; WHO standard: 6.5-8.5.               | Numerical   |
| Hardness        | Indicates concentration of calcium and magnesium salts.                       | Numerical   |
| TDS             | Total dissolved substances; max limit: 1000 mg/l.                             | Numerical   |
| Chloramines     | Primary disinfectant in water; safe up to 4 mg/L.                             | Numerical   |
| Sulfate         | Naturally occurring in minerals; typical level: 3-30 mg/L.                    | Numerical   |
| Conductivity    | Measures water's ion concentration; max limit: 400 µS/cm.                     | Numerical   |
| Organic Carbon  | Total carbon in organic compounds; EPA limit: below 2 mg/L.                   | Numerical   |
| Trihalomethanes | Byproducts from chlorine; safe up to 80 ppm.                                  | Numerical   |
| Turbidity       | Gauge of suspended solids; WHO limit: 5.00 NTU.                               | Numerical   |
| Potability      | Consumption indicator whether water is safe or not; 1 for potable, 0 for not. | Categorical |

TABLE II  
STATISTICAL SUMMARY OF DATASET ATTRIBUTES

| Attribute       | Count | Mean         | Std         | Min        | 25%          | 50%          | 75%          | Max          |
|-----------------|-------|--------------|-------------|------------|--------------|--------------|--------------|--------------|
| pH              | 3276  | 7.080855     | 1.469958    | 0          | 6.277673     | 7.085378     | 7.870050     | 14           |
| Hardness        | 3276  | 196.369496   | 32.879761   | 47.432000  | 176.850538   | 196.967627   | 216.667456   | 323.124000   |
| Solids          | 3276  | 22014.092526 | 8768.570828 | 320.942611 | 15666.690297 | 20927.833607 | 27332.762127 | 61227.196008 |
| Chloramines     | 3276  | 7.122277     | 1.583085    | 0.352000   | 6.127421     | 7.130299     | 8.114887     | 13.127000    |
| Sulfate         | 3276  | 333.785123   | 36.145701   | 129        | 317.094638   | 334.564290   | 350.385756   | 481.030642   |
| Conductivity    | 3276  | 426.205111   | 80.824064   | 181.483754 | 365.734414   | 421.884968   | 481.792304   | 753.342620   |
| Organic Carbon  | 3276  | 14.284970    | 3.308162    | 2.200000   | 12.065801    | 14.218338    | 16.557652    | 28.300000    |
| Trihalomethanes | 3276  | 66.395671    | 15.769901   | 0.738000   | 56.647656    | 66.303555    | 76.666609    | 124          |
| Turbidity       | 3276  | 3.966786     | 0.780382    | 1.450000   | 3.439711     | 3.955028     | 4.500320     | 6.739000     |

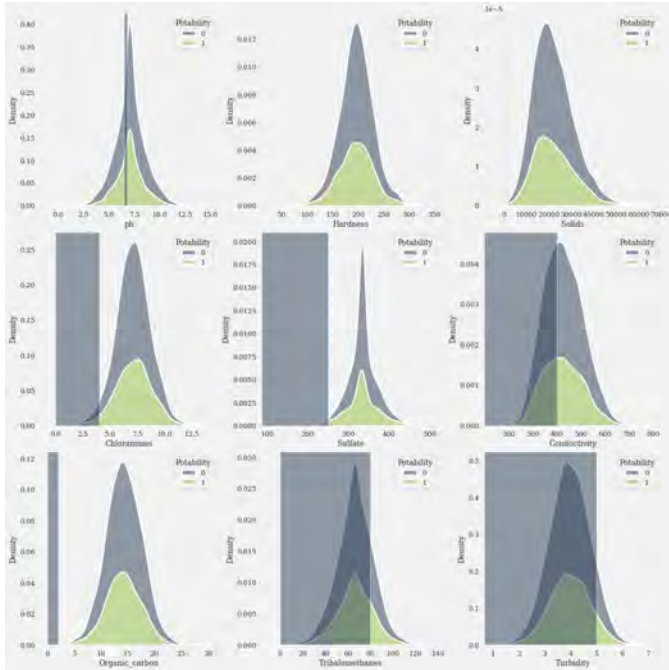


Fig. 1. Distribution of other features relative to the potability feature within the dataset

### C. Predictive Model Development

After evaluating various state-of-the-art models, a Hyperparameter-Tuned Histogram Gradient Boosting Classifier model was developed to further improve the classification performance of water quality.

A Histogram Gradient Boosting Classifier combines the

principles of gradient boosting with histogram-based techniques to enhance efficiency and performance in training predictive models for classification tasks. This classifier works in three steps, namely, Initialization, Iterative Training, and Histogram Construction. The steps are provided below.

#### 1) Initialization:

- Begin with an initial model, typically a simple one like a single decision stump.
- Initialize the model with a constant prediction, often the mean of the target variable:

$$\hat{y}^0(x) = \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$$

where  $\hat{y}^0(x)$  is the initial prediction for all samples  $x$ ,  $\bar{y}$  is the mean of the target variable  $y$ , and  $N$  is the number of samples.

#### 2) Iterative Training:

- For each boosting iteration  $m$ , fit a new decision tree  $h_m(x)$  to the negative gradient of the loss function  $L(y, \hat{y}^{m-1}(x))$ :
  - Compute the negative gradient:

$$\text{Gradient}_i = -\frac{\partial L(y_i, \hat{y}^{m-1}(x_i))}{\partial \hat{y}^{m-1}(x_i)}$$

- Fit the decision tree:

$$h_m(x) = \arg \min_h \sum_{i=1}^N L(y_i, \hat{y}^{m-1}(x_i) + h(x_i))$$

- b) Update the model by adding the new decision tree, weighted by a learning rate  $\eta$ :

$$\hat{y}^m(x) = \hat{y}^{m-1}(x) + \eta h_m(x)$$

where  $\hat{y}^m(x)$  is the updated prediction after  $m$  iterations.

3) Histogram Construction:

- Instead of evaluating all possible splits for each feature, histogram-based methods discretize continuous features into bins (histograms) and calculate histograms of gradients for efficient split finding:

$$\text{Histogram} = \sum_{i=1}^N (\text{Gradient}(y_i, \hat{y}^{m-1}(x_i)))$$

This reduces the computational complexity significantly compared to evaluating every split, making it suitable for large datasets.

After constructing the base model, hyperparameters were optimized to enhance performance. Hyperparameter tuning was conducted employing Grid Search method, which systematically explores a predefined subset of hyperparameters. This method defines a grid or range of hyperparameter values and evaluates the model efficacy for each composition. The specific hyperparameter grid utilized in this study is detailed in Table III. Table IV showcases the optimal parameters identified and utilized in this study. Algorithm 1 outlines the procedural steps employed to construct the Histogram Gradient Boosting Classifier.

TABLE III  
PARAMETER GRID FOR MODEL TUNING

| Parameter           | Values                  |
|---------------------|-------------------------|
| maximum iteration   | 100 - 300 (step in 100) |
| maximum depth       | Nan, 5, 10              |
| minimum sample leaf | 1, 2, 4                 |
| l2_regularization   | 0.1, 1.0, 10.0          |
| maximum bins        | 32, 64, 128             |

TABLE IV  
BEST PARAMETERS TO CONSTRUCT THE HISTOGRAM GRADIENT BOOSTING CLASSIFIER

| Parameter           | Value |
|---------------------|-------|
| l2_regularization   | 10.0  |
| maximum bins        | 128   |
| maximum depth       | 10    |
| maximum iteration   | 100   |
| minimum sample leaf | 2     |

The histogram gradient boosting algorithm has advantages over traditional machine learning models. Primarily, it enhances efficiency by employing histograms of gradients, which streamline the process of identifying optimal feature splits during training. This efficiency improvement translates into faster training times, making it particularly advantageous for handling large datasets. Moreover, the algorithm's reduced computational overhead and lower memory usage compared

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**Algorithm 1:** Histogram Gradient Boosting Classifier with Hyperparameter Tuning

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**Input:** Dataset  $D$

**Output:** Predicted class labels  $\hat{y}_i$  for all samples  $x_i$

**Initialization:**

Initialize with a constant prediction:

$$\hat{y}^0(x) = \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$$

**Boosting Iterations:**

**for**  $m = 1$  **to**  $M$  **do**

Calculate the negative slope or gradient of the loss function:

$$\text{Gradient}_i = -\frac{\partial L(y_i, \hat{y}^{m-1}(x_i))}{\partial \hat{y}^{m-1}(x_i)}$$

Construct histograms of gradients for feature bins:

$$\text{Histogram} = \sum_{i=1}^N (\text{Gradient}(y_i, \hat{y}^{m-1}(x_i)))$$

Fit a decision tree to minimize the loss:

$$h_m(x) = \arg \min_h \sum_{i=1}^N L(y_i, \hat{y}^{m-1}(x_i) + h(x_i))$$

Update the model with a learning rate  $\eta$ :

$$\hat{y}^m(x) = \hat{y}^{m-1}(x) + \eta h_m(x)$$

**end**

**Hyperparameter Tuning:**

Tune hyperparameters  $\eta$ , max depth of trees, and number of boosted iterations  $M$  using cross-validation:

$$\eta^*, M^* = \arg \max_{\eta, M} \text{Cross-validation score}$$


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to exhaustive search methods contribute to its scalability, allowing it to effectively manage substantial volumes of data. In terms of predictive performance, histogram gradient boosting often achieves higher accuracy by effectively capturing complex data relationships without succumbing to overfitting, thus enhancing its utility across various predictive modeling tasks.

#### D. Experimental Setup and Evaluation Metrics

1) *Experimental Setup:* The experiments were conducted using Google Colab and the Kaggle notebook platform, accessed from a local machine running Windows 11 OS and utilizing the Edge browser. The local machine configuration consisted of an Intel Core i3 3rd generation laptop with 12GB RAM and a 128GB SSD. All the experimental codes were written using python programming language version 3.11 and scikit-learn library version 1.5.

2) *Evaluation Metrics*: The models were evaluated on the standard classification metrics namely, Accuracy, ROC-AUC and F1 Score.

a) *Accuracy*:

- **Definition**: Accuracy computes the proportion of correctly classified cases among the total cases in a dataset.
- **Formula**:

$$\text{Accuracy} = \frac{\text{Number of correct forecast}}{\text{Total number of forecast}}$$

b) *ROC-AUC (Receiver Operating Characteristic - Area Under the Curve)*:

- **Definition**: ROC-AUC provides a comprehensive measure of the model's discriminatory power, capturing the trade-off between the true positive rate (sensitivity) and the false positive rate across varying thresholds.
- **Formula**:

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt$$

Where TPR stands for True Positive Rate and FPR stands for False Positive Rate.

c) *F1 Score*:

- **Definition**: F1 score is the harmonic mean of precision and recall metrics.
- **Formula**:

$$F_1 = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}}$$

### III. RESULTS AND DISCUSSION

This section presents the experimental results achieved from the experiment. The results of the evaluation metrics for the hyper tuned histogram gradient boosting model is shown in Table V.

TABLE V  
CLASSIFICATION REPORT OF THE HYPER TUNED HISTOGRAM GRADIENT BOOSTING MODEL

| Class                   | Precision | Recall | F1  | Support |
|-------------------------|-----------|--------|-----|---------|
| 0                       | 69%       | 80%    | 74% | 250     |
| 1                       | 60%       | 46%    | 52% | 163     |
| <b>Accuracy</b>         | 67%       |        |     |         |
| <b>Macro Average</b>    | 65%       | 63%    | 63% | 413     |
| <b>Weighted Average</b> | 66%       | 67%    | 66% | 413     |

Table V presents the classification performance of the hyper tuned histogram gradient boosting model, evaluated on a test set with 413 samples over 5-fold cross validation. For class 0, the model achieved a precision of 69%, a recall of 80%, and an F1-score of 74%, indicating a relatively good balance between precision and recall. For class 1, precision was 60%, recall was 46%, and the F1-score was 52%, suggesting a lower performance in detecting this class compared to class 0. Overall, the model achieved an accuracy of 67%. The macro average values for precision, recall, and F1-score were 65%, 63%, and 63%, respectively, providing an unweighted mean performance across both classes. The weighted average values

were 0.66 for precision, 0.67 for recall, and 0.66 for F1-score, accounting for the support (number of instances) of each class. This indicates that the model's performance is skewed towards class 0, which has more instances in the test set. The ROC-AUC score provides an aggregate measure of performance across all classification thresholds, with a score of 0.7034 indicating that the model has a 70.34% chance of correctly differentiating between a erratically chosen positive instance and a erratically chosen negative instance. Figure 2 presents the ROC-AUC curve for the hyper tuned histogram gradient boosting model.

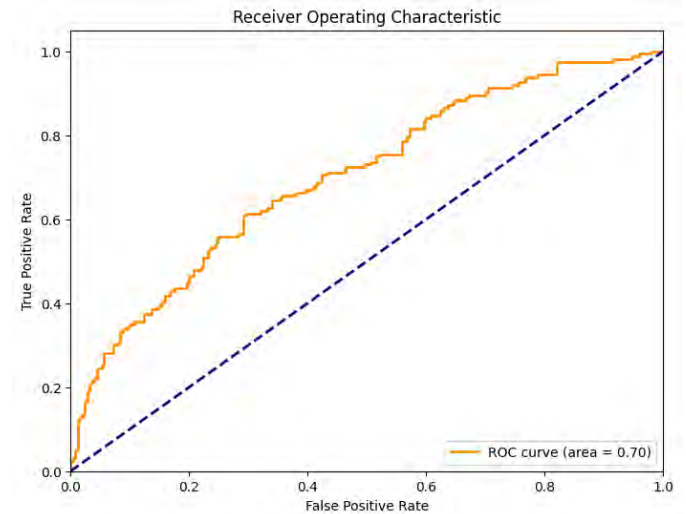


Fig. 2. ROC-AUC curve for the hyper tuned histogram gradient boosting model

Besides the hyper tuned histogram gradient boosting model, the dataset was also evaluated with a good number of SOTA ML models. Table VI presents the comparison of various ML models in terms of accuracy, ROC-AUC, and F1-score. The models range from classic algorithms like Random Forest Classifier and SVC to advanced deep learning models like BiLSTM and BiGRU. Among these, the Hyper Tuned Histogram Gradient Boosting model demonstrated better performance with an accuracy of 67%, a ROC-AUC score of 70%, and an F1-score of 66%, indicating its ability to effectively balance between precision and recall, and provide a robust classification performance. Other models such as NuSVC, XGB Classifier, and Random Forest Classifier also performed well but with slightly lower scores. The results highlight the significance of model choice and hyperparameter tuning in achieving optimal performance for classification tasks related to water potability.

TABLE VI  
PERFORMANCE METRICS OF DIFFERENT MODELS

| Model                                          | Accuracy    | ROC-AUC     | F1-Score    |
|------------------------------------------------|-------------|-------------|-------------|
| NuSVC                                          | 0.65        | 0.62        | 0.64        |
| XGB Classifier                                 | 0.63        | 0.61        | 0.62        |
| Random Forest Classifier                       | 0.64        | 0.60        | 0.61        |
| Quadratic Discriminant Analysis                | 0.64        | 0.59        | 0.61        |
| Extra Trees Classifier                         | 0.64        | 0.59        | 0.60        |
| SVC                                            | 0.64        | 0.59        | 0.60        |
| K Neighbors Classifier                         | 0.62        | 0.58        | 0.60        |
| Label Propagation                              | 0.60        | 0.57        | 0.59        |
| Label Spreading                                | 0.60        | 0.57        | 0.59        |
| Bagging Classifier                             | 0.61        | 0.57        | 0.59        |
| Decision Tree Classifier                       | 0.58        | 0.56        | 0.58        |
| Gaussian NB                                    | 0.60        | 0.54        | 0.54        |
| SGD Classifier                                 | 0.60        | 0.52        | 0.50        |
| Ada Boost Classifier                           | 0.56        | 0.51        | 0.52        |
| Passive Aggressive Classifier                  | 0.53        | 0.50        | 0.52        |
| Logistic Regression                            | 0.59        | 0.50        | 0.44        |
| Linear Discriminant Analysis                   | 0.59        | 0.50        | 0.44        |
| Ridge Classifier                               | 0.59        | 0.50        | 0.44        |
| Ridge Classifier CV                            | 0.59        | 0.50        | 0.44        |
| Dummy Classifier                               | 0.59        | 0.50        | 0.44        |
| Calibrated Classifier CV                       | 0.59        | 0.50        | 0.44        |
| Bernoulli NB                                   | 0.59        | 0.50        | 0.44        |
| Linear SVC                                     | 0.59        | 0.50        | 0.44        |
| Nearest Centroid                               | 0.50        | 0.50        | 0.50        |
| Perceptron                                     | 0.46        | 0.48        | 0.47        |
| BiLSTM                                         | 0.58        | 0.50        | 0.58        |
| BiGRU                                          | 0.58        | 0.50        | 0.59        |
| <b>Hyper Tuned Histogram Gradient Boosting</b> | <b>0.67</b> | <b>0.70</b> | <b>0.66</b> |

#### IV. CONCLUSION

This work demonstrated the effectiveness of ML techniques, particularly a Hyperparameter-Tuned Histogram Gradient Boosting Classifier, in predicting water potability. The model achieved an accuracy of 67%, a ROC-AUC score of 70%, and an F1-score of 0.66, outperforming several other ML algorithms. Achieved results highlight the ability of advanced ML techniques in water quality assessment and management. However, the study has some limitations. The model's performance, while promising, still leaves room for improvement. The class imbalance issue, even after applying SMOTE, may have adverse effect in predicting the minority class accurately. Future research directions may include exploring techniques like ensemble methods incorporating temporal and spatial data to capture dynamic changes in water quality. Multi source dataset and real time dataset can also be investigated to improve prediction accuracy. Investigating the interpretability of the model to provide perceptions into the most influential factors affecting water potability would also be valuable using explainable AI. Furthermore, developing an instantaneous water quality monitoring system based on the predictive model could have practical applications in water management. By tackling these limitations and following these future directions, researchers can further augment the accuracy and pertinency of ML models in water quality prediction, ultimately contributing to improved water resource management and public health outcomes.

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